Question Paper

Exam Date & Time: 25-Nov-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: CSS 2103/CSS_2103 DATA ANALYTICS
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) The Glossary manager wants to boost the basket value and reduce missed cross-sell opportunities. The manager believes that certain items, such as milk, bread, cheese, butter, eggs, bananas, and coffee, are frequently bought together; these items should be co-located on the shelves. For this scenario, apply the Apriori algorithm to verify that buying patterns are interesting to the users. Consider the following transaction logs shown in Table Q.1A (5)
Answer the followings(candidate set represented as C, Frequent itemset represented as L):

Table Q.1A

TID	Items
T1	milk, bread, egg
T2	milk, bread, banana
T3	milk, cheese, banana
T4	bread, butter, cheese
T5	milk, bread, butter, cheese
T6	milk, coffee, bread
T7	banana, cheese
T8	milk, bread, egg, butter
T9	milk, bread, cheese
T10	coffee, bread

i. Build $C_1 \rightarrow L_1$, $C_2 \rightarrow L_2$, $C_3 \rightarrow L_3$ with min support count = 4; list pruned candidates at each stage.
ii. Generate rules from the largest L_k with min_conf = 70%;
Compute support, confidence. If min_sup changes to 5, which itemsets drop? Briefly explain via the Apriori property.

Course Outcome: "CO4" / Cognitive Level: "Application"

1B) A medical equipment company claims that its new blood test kit has an average accuracy rate of **95%**. A quality inspector believes the true accuracy **may be different** from the claimed value (it could be higher or lower). To verify the claim, a random sample of **36 test kits** is taken, and the sample shows an **average accuracy of 94.7%** with a **known population standard deviation of 0.02**. At the **5% significance level**, test whether the company's claim is justified. (critical z-values are ±**1.96)** (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

1C) An ice cream manufacturer noticed that sales of ice cream increase every summer and drop in winter. The sales pattern repeats each year with increasing average sales overtime due to marketing efforts. Which components are present in this time series? Identify the time series model that would best represent this pattern and why? (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

2A) Table Q. 2A shown below has the data collected during the outbreak of smallpox. (5)

Table Q.2A

	Attacked	Not attacked	Total
Vaccinated	31	469	500
Not Vaccinated	185	1315	1500
Total	216	1784	2000

Test the effectiveness of vaccination in preventing smallpox at 5% significance level.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

2B) Consider frequent 1-itemsets: (3)
L_1 = {A, B, C, D, E, F}.
Use the hash function

$$h(x, y) = ((order(x) \times 5) + order(y)) \bmod 7$$

and the following transactions:
T_1: {A, B, C}
T_2: {A, D, E}
T_3: {B, C, E, F}
T_4: {A, B, E, F}.
i. Construct the hash table H_2 showing bucket addresses, counts, and contents.
ii. If the minimum support is 4, which buckets can be pruned?

Course Outcome: "CO4" / Cognitive Level: "Analysis"

2C) Two forecasting models for production planning are evaluated as follows: (2)

Model	MAE	MSE	RMSE	MAPE
Model A	12	25	5	6%
Model B	10	36	6	4%

Justify and explain which model gives better overall forecasting performance? Explain, considering all measures

Course Outcome: "CO5" / Cognitive Level: "Application"

3A) i.Find the standard deviation and skewness for the following data: (5)
78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75 79, 63, 42, 61
ii. Give example for "Combining variables" concept in dataset preparation.
iii. A CMO of a company marketing carbonated soft drinks has developed three new flavors of cola and decided to measure the customers' preference for these three flavors from 12 customers by using scale of 1 to 9. The table of data is shown below. Conduct the appropriate analysis here.

Flavor-A	Flavor-B	Flavor-C
4	6	8
2	4	6
4	5	8
2	5	6

Course Outcome: "CO1" / Cognitive Level: "Application"

3B) A dataset contains 100 transactions. (3)
• 20 transactions include both {Pen, Notebook}.
• 25 transactions include {Pen}.
• 40 transactions include {Notebook}.
i. Calculate **Support** and **Confidence** for the rule {Pen} → {Notebook}.
ii. If the minimum support is 10% and minimum confidence is 60%, check whether the rule is **strong**.
iii. Interpret the result.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

3C) Consider the sales for the past 5 weeks: 120, 130, 125, 140, 150. (2)
i. Compute the 3-week Simple Moving Average (SMA) forecast for week 6.
ii. Comment briefly on the difference between the SMA and WMA forecasts.

Course Outcome: "CO5" / Cognitive Level: "Application"

4A) Given the following transaction database: (5)

TID	Items
1	A, C
2	A, B, D
3	A, B, C
4	B, C, D
5	A, C

a) Convert the above database into vertical data format and find all frequent itemsets with a minimum support of 2.
b) List the advantages of using vertical data format over horizontal format in frequent itemset mining?

Course Outcome: "CO4" / Cognitive Level: "Application"

4B) A company tracks monthly sales (in units) for a product: (3)

Month	Sales
1	120
2	135
3	150
4	160
5	170

Using **Single Exponential Smoothing** (SES) with a smoothing constant **$\alpha = 0.3$**, and assuming the forecast for Month 1 is **120 units**,
(a) Calculate the forecast for Months 2 to 6.
(b) Compute the **Mean Absolute Error (MAE)** for the forecasts from Months 2 to 5.

Course Outcome: "CO5" / Cognitive Level: "Application"

4C) A supermarket's market-basket analysis generates several association rules, such as: (2)
• Rule 1: Diapers → Beer (support = 5%, confidence = 80%)
• Rule 2: Milk → Bread (support = 20%, confidence = 60%)
Some rules may appear strong based on support and confidence but can be misleading.
a. Explain why some association rules can be misleading even if they have high support and confidence.
b. Identify at least two additional measures that can be used to evaluate the true strength of a rule.

Course Outcome: "CO4" / Cognitive Level: "Understand"

5A) A transportation company tracks its monthly passenger counts (Y_t). The data for the first five months are given. Applying Holt's Method (5)

Period	Month	Passenger Count
1	Jan	400
2	Feb	450
3	Mar	520
4	Apr	500
5	May	550

Parameters: Use smoothing constants
a. Calculate the initial Level (L_1) and Trend (T_2).
b. Calculate the final Level (L_5) and Trend (T_5) based on all five data points.
c. Use L_5 and T_5 to provide the forecast for Month 7 (F_7).

Course Outcome: "CO5" / Cognitive Level: "Application"

5B) An online electronics store analysed 1,000 customer purchase transactions. The following data were obtained: (3)

Itemset	Number of Transactions
{Smartphone}	420
{Wireless Earbuds}	380
{Smartphone, Wireless Earbuds}	260

(a) Calculate the pattern evaluation measures such as Lift, Kulc and cosine for the following association rule:
Smartphone → WirelessEarbuds
(b) Justify the relationship between Smartphone and Earbud purchases based on the obtained Lift and Kulc values.

Course Outcome: "CO4" / Cognitive Level: "Application"

5C) A **data analyst** is studying the **annual income levels (in ₹000s)** of 12 employees in a company: (2)
[180, 200, 210, 215, 220, 230, 250, 270, 300, 320, 400, 480]
Apply both **equal-width** and **equal-frequency binning** with **3 bins**.
Using **histogram analysis,** discuss which binning approach (**equal-width** or **equal-frequency)** provides a **better representation** of the income distribution and justify your reasoning.

Course Outcome: "CO3" / Cognitive Level: "Application"

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Event Name	Subject
END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2103:DATA ANALYTICS

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	3	CO 3	10
3	4	CO 4	21
4	5	CO 5	14
		Total	**50**

* NA - CO Not Selected

Event Name	Subject
END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2103:DATA ANALYTICS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	11	37(74%)
4	Analysis	3	11(22%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)